

GOVT 6019: Homework 2 — Solutions

Fall 2026 · Describing data

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Setup used throughout (the path is relative to this file's folder in the course repository; adjust to your own project's layout):

```
library(tidyverse)
source("../scripts/gov6019_ggplot.R") # theme_gov() and course colors
voter <- read_csv("../data/voter2012.csv")
```

Part A — The algebra of center and spread

Question 1: The balancing point

(a) Split the sum, then use the definition $\bar{X} = \frac{1}{n} \sum_i X_i$, i.e., $\sum_i X_i = n\bar{X}$:

$$\sum_{i=1}^n (X_i - \bar{X}) = \sum_{i=1}^n X_i - \sum_{i=1}^n \bar{X} = n\bar{X} - n\bar{X} = 0.$$

The only facts used: a sum splits across subtraction, and adding the constant $\bar{X}$ to itself n times gives $n\bar{X}$.

(b) The same two moves, with a general constant c :

$$\sum_{i=1}^n (X_i - c) = \sum_{i=1}^n X_i - nc = n\bar{X} - nc = n(\bar{X} - c).$$

Since $n \geq 1$, the right-hand side is zero **if and only if** $\bar{X} - c = 0$, i.e., $c = \bar{X}$. The mean is therefore the **unique balancing point** of the data: every other choice of c leaves an imbalance of exactly $n(\bar{X} - c)$.

(c) By (a), the average deviation $\frac{1}{n} \sum_i (X_i - \bar{X})$ equals zero for *every* dataset — for five counties clustered at 50 and for five counties scattered from 30 to 70 alike — so it carries no information about spread at all. The two repairs from lecture: take absolute values before averaging (the mean absolute deviation) or square the deviations before averaging (the variance, and its square root the SD).

Question 2: Rescaling without fear

(a) Push the sum through $Y_i = a + bX_i$:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n (a + bX_i) = \frac{1}{n} \left(na + b \sum_{i=1}^n X_i \right) = a + b\bar{X}.$$

The mean rides along with any linear transformation.

(b) First note what the transformation does to deviations. Using (a),

$$Y_i - \bar{Y} = (a + bX_i) - (a + b\bar{X}) = b(X_i - \bar{X}),$$

so every deviation is simply scaled by b — the shift a has already canceled. Then

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 = \frac{1}{n-1} \sum_{i=1}^n b^2 (X_i - \bar{X})^2 = b^2 s_X^2, \quad s_Y = \sqrt{b^2 s_X^2} = |b| s_X.$$

Why did a disappear? Adding a constant slides every observation **and** the mean by the same amount, so no observation's distance to the mean changes — location moves, spread does not.

Why the absolute value? Because $\sqrt{b^2} = |b|$, not b . A negative b flips the ordering of the data (the largest X becomes the smallest Y), but distances between points are unchanged, and an SD cannot be negative.

(c) The rescale is

$$\frac{4-x}{3} = \frac{4}{3} + \left(-\frac{1}{3}\right)x, \quad \text{so} \quad a = \frac{4}{3}, \quad b = -\frac{1}{3}.$$

Raw item, Don't knows removed:

```
gayr <- na_if(voter$important_gayr, 5)
c(mean = mean(gayr, na.rm = TRUE), sd = sd(gayr, na.rm = TRUE))
```

```
      mean      sd
2.676175 1.115783
```

Predictions from the formulas — mean $a + b\bar{X} = (4 - \bar{X})/3$ and SD $|b| s_X = s_X/3$:

```
c(pred_mean = (4 - mean(gayr, na.rm = TRUE)) / 3,
  pred_sd    = sd(gayr, na.rm = TRUE) / 3)
```

```
pred_mean  pred_sd
0.4412751  0.3719276
```

Now rescale and check:

```
gayr01 <- (4 - gayr) / 3
c(mean = mean(gayr01, na.rm = TRUE), sd = sd(gayr01, na.rm = TRUE))
```

```
      mean      sd
0.4412751 0.3719276
```

The predictions hit to every printed decimal: raw mean 2.676 and SD 1.116 become mean 0.441 and SD 0.372. No re-computation from the microdata was ever needed — that is the point of the algebra.

(d) The SD is measured in the variable's *units*, and the rescale changed the units: dividing the scale by 3 divides the SD by 3, mechanically, for any data whatsoever. Relative to the length of its scale, the rescaled item is exactly as dispersed as the original — no noise was removed, only relabeled. (Comparing dispersion across differently-scaled variables needs standardizing, which arrives in Week 4.)

Question 3: The mean minimizes squared deviations

(a) Add and subtract $\bar{X}$ inside the square and expand $(u + v)^2 = u^2 + 2uv + v^2$ with $u = X_i - \bar{X}$ and $v = \bar{X} - c$:

$$\begin{aligned} g(c) &= \sum_{i=1}^n (X_i - c)^2 = \sum_{i=1}^n \left[(X_i - \bar{X}) + (\bar{X} - c) \right]^2 \\ &= \sum_{i=1}^n (X_i - \bar{X})^2 + 2(\bar{X} - c) \sum_{i=1}^n (X_i - \bar{X}) + n(\bar{X} - c)^2. \end{aligned}$$

(The middle term: $(\bar{X} - c)$ is a constant, so it pulls out of the sum; the last term: $(\bar{X} - c)^2$ summed n times.)

(b) The middle term contains $\sum_i (X_i - \bar{X})$, which is zero by Question 1(a), so the cross-term vanishes.

(c) What remains is

$$g(c) = \underbrace{\sum_{i=1}^n (X_i - \bar{X})^2}_{\text{does not involve } c} + \underbrace{n(\bar{X} - c)^2}_{\geq 0}.$$

The first term is fixed; the second is a squared quantity times n , so it is nonnegative and equals zero **only** when $c = \bar{X}$. Any other choice of c pays a strictly positive penalty of $n(\bar{X} - c)^2$. Hence g is minimized at $c = \bar{X}$, uniquely, and the minimum value is

$$g(\bar{X}) = \sum_{i=1}^n (X_i - \bar{X})^2 = (n - 1) s^2.$$

The smallest achievable sum of squared deviations *is* (up to the $n - 1$ divisor) the sample variance: s^2 measures the spread that remains after the best possible single-number summary has done its work.

(d) **Calculus aside** (not required). Differentiate g with respect to c :

$$g'(c) = \sum_{i=1}^n 2(X_i - c)(-1) = -2 \sum_{i=1}^n (X_i - c) = -2n(\bar{X} - c),$$

where the last step is Question 1(b). Setting $g'(c) = 0$ gives $c = \bar{X}$, and $g''(c) = 2n > 0$ confirms a minimum. Same answer, two lines — but the algebraic route in (a)–(c) shows *more*: it gives the exact size of the penalty for any wrong c , not just the location of the best one.

Remark. The companion fact: the **median** minimizes the sum of *absolute* deviations $\sum_i |X_i - c|$ (the absolute value’s kink at zero is what makes that proof harder, and is the same kink that pushed statistics toward squares). And choosing a summary by minimizing squared distance is precisely the move that will produce the regression line in Week 10 — “least squares” is this question with more variables.

Part B — Measurement

Question 4: Reliability or validity?

- a. **Reliability — inter-coder.** The same instrument (codebook) applied to the same events by two measurers returns different answers 39% of the time; with only two categories, pure coin-flipping would already agree about half the time, so 61% is alarming. The measurement varies with the measurer, and no downstream use of the codes can outrun that noise — unreliability caps validity. *Improvement:* sharpen the codebook with explicit decision rules and worked borderline examples, retrain, have both RAs double-code a subsample, adjudicate disagreements, and report the agreement rate alongside the data.
- b. **Validity — criterion.** The measure is demonstrably *reliable* — people repeat the same answer months later — but against the official-record benchmark it is systematically wrong by 14 points, and in a known direction (voting is a civic norm, so overclaiming dominates). Consistency is no defense: respondents reliably report their civic self-image, which is not the behavior we wanted. *Improvement:* use validated voter-file matches where possible, or question wordings that give respondents a face-saving way to admit not voting.
- c. **Validity — construct.** Stable scores certify reliability, but the critic’s charge is that the quiz measures a *neighboring construct* — political knowledge, itself heavily a product of schooling and news exposure — rather than interest. Someone can be intensely interested and poorly informed, or informed and bored. *Improvement:* measure interest directly (self-reported attention and engagement items), and check that the quiz and the direct items co-move before treating either as the construct.
- d. **Validity — face (and construct).** The rule equates democracy with holding multiparty elections, but regimes can stage regular elections while jailing opponents and muzzling the press — the measure misses most of what the concept means on its face. Note the pairing with (a): here coders “almost never disagree,” so reliability is superb precisely *because* the rule is mechanical — a perfectly reliable ruler, consistently measuring the wrong thing. *Improvement:* code the further dimensions (civil liberties, press freedom, opposition rights) and combine them — which immediately raises this week’s index-construction questions; there is no free lunch in measurement.

Two morals: reliability and validity are separate audits, and **a measure can pass one perfectly while failing the other** — (b) and (d) are reliable and invalid, while (a) is unreliable before validity can even be assessed.

Question 5: Defend or attack this index

A model memo (yours need not match, but it should hit problems of each kind):

Three problems, in ascending order of severity. **First — disguised nonresponse (fixable).** `pol_interest` codes “Not sure” as 4, *above* “Very much” (3), and

`important_econ` codes “Don’t know” as 5; as proposed, the index scores maximal uncertainty as maximal engagement. `na_if()` both before any arithmetic. **Second — direction and scale (fixable).** `important_econ` runs *backwards* relative to the other two (1 = strongly agree the economy is important), so the components pull in opposite directions; and the raw ranges (1–4, 1–6, 1–5) are unequal, so `educ`’s six-point range quietly dominates the variation — an implicit weighting nobody chose. Flip the reversed item and rescale all three to 0–1 before averaging. **Third — construct validity (fatal, in my view).** Even fully cleaned, these three items measure three different things: interest in politics (an engagement-adjacent attitude), educational attainment (a resource that *correlates* with engagement), and the belief that the economy is an important issue (an attitude about issues, not about participating). An additive index borrows its logic from items sharing one underlying signal; these do not. **Verdict: abandon.** With only this file, cleaned `pol_interest` is the one face-valid engagement item we have — report it honestly on its own, and keep issue importance as the separate construct it is. Behavioral engagement (voting, donating, attending) is simply not in these columns.

“Repair” was also a defensible verdict *if* the memo both fixed the coding problems concretely (recode, flip, rescale) and renamed the result for what it then measures — something like “politically attentive resources” — while conceding the construct objection. What does not work: memos that treated the recoding hazards as the only problems, or that attacked the very idea of indices (the index you built in lab and the lecture’s item-noise argument say otherwise).

Part C — Applied: the social-issues importance index

Question 6: Build it

(a) Count the Don’t knows and audit the recode. New columns, per lab’s rule — the raw ones stay untouched as evidence (and Question 7 needs them):

```
voter |>
  summarise(
    dk_gayr   = sum(important_gayr == 5, na.rm = TRUE),
    dk_abort  = sum(unimportant_abort == 5, na.rm = TRUE),
    na_gayr   = sum(is.na(important_gayr)),
    na_abort  = sum(is.na(unimportant_abort))
  )
```

```
# A tibble: 1 x 4
  dk_gayr dk_abort na_gayr na_abort
  <int>    <int>    <int>    <int>
1     39      39     108     106
```

```
voter <- voter |>
  mutate(
    gayr_c   = na_if(important_gayr, 5),
    abort_c  = na_if(unimportant_abort, 5)
  )
```

```
voter |>
```

```
summarise(na_gayr_after = sum(is.na(gayr_c)),
          na_abort_after = sum(is.na(abort_c)))
```

A tibble: 1 x 2

```
na_gayr_after na_abort_after
      <int>         <int>
1      147         145
```

Each item had **39** Don't knows. The books balance exactly: `important_gayr` goes from 108 missing to $108 + 39 = 147$, and `unimportant_abort` from 106 to $106 + 39 = 145$. If your before/after difference is not exactly the number of 5s, the recode did something you did not intend.

(b) Flip the reverse-framed abortion item, *after* DK removal — into a new column whose name states its new direction:

```
voter <- voter |> mutate(abort_f = 5 - abort_c)

table(raw = voter$unimportant_abort, flipped = voter$abort_f,
      useNA = "ifany")
```

	flipped				
raw	1	2	3	4	<NA>
1	0	0	0	1310	0
2	0	0	1833	0	0
3	0	2280	0	0	0
4	2432	0	0	0	0
5	0	0	0	0	39
<NA>	0	0	0	0	106

The order matters. Because $5 - 5 = 0$, flipping first turns every Don't know into a phantom value of 0 — below the scale, invisible to a later `na_if(x, 5)` (there are no 5s left to find), and silently included in every summary afterward. Order of operations is part of the pipeline's correctness.

The crosstab shows everything on the anti-diagonal ($1 \leftrightarrow 4$, $2 \leftrightarrow 3$) with the 39 DKs and 106 genuine blanks in the NA column. The **2,432** respondents who *strongly disagreed* that abortion is unimportant (raw 4) now sit at flipped value 1 — which the rescale in (c) sends to the top of the importance scale, where they belong.

(c) Rescale to 0–1 and average the two items:

```
voter <- voter |>
  mutate(
    gayr_01 = (4 - gayr_c) / 3,
    abort_01 = (4 - abort_f) / 3
  )

voter$social_index <- voter |>
  select(gayr_01, abort_01) |>
  rowMeans(na.rm = TRUE)
voter$social_index[is.nan(voter$social_index)] <- NA # answered neither item
```

```
voter |>
  summarise(scored = sum(!is.na(social_index)),
            mean   = mean(social_index, na.rm = TRUE),
            median = median(social_index, na.rm = TRUE),
            sd     = sd(social_index, na.rm = TRUE))
```

```
# A tibble: 1 x 4
  scored mean median   sd
  <int> <dbl> <dbl> <dbl>
1   7917 0.511   0.5 0.286
```

7,917 of the 8,000 respondents receive a score; the index has mean **0.51**, median **0.50**, and SD **0.29**.

(d) Available-case versus listwise:

```
voter <- voter |>
  mutate(social_index_lw = (gayr_01 + abort_01) / 2) # NA if either missing
```

```
voter |>
  summarise(
    available_case = sum(!is.na(social_index)),
    listwise       = sum(!is.na(social_index_lw)),
    one_item_only  = sum(!is.na(social_index) & is.na(social_index_lw)),
    never_scored   = sum(is.na(social_index))
  )
```

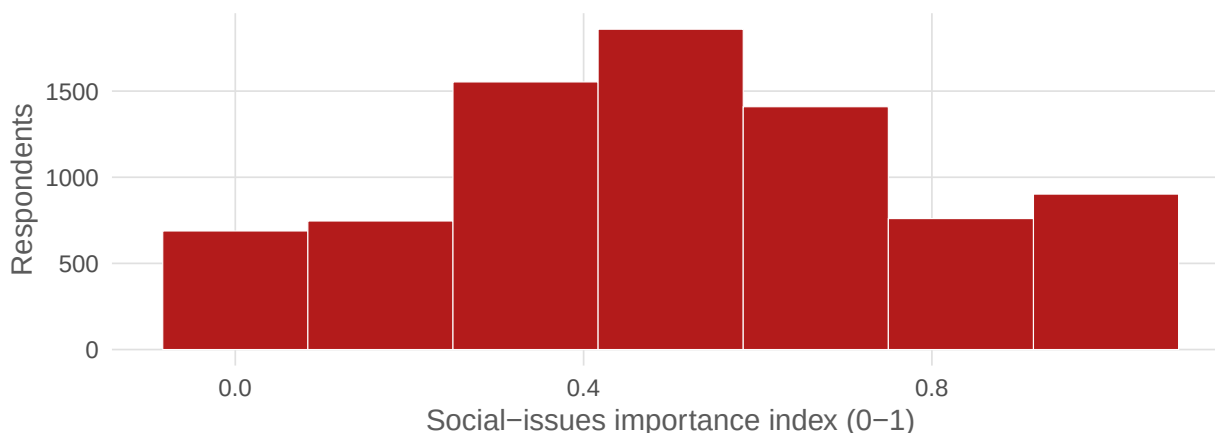
```
# A tibble: 1 x 4
  available_case listwise one_item_only never_scored
      <int>      <int>      <int>      <int>
1      7917      7791      126        83
```

126 respondents answered exactly one of the two items: `na.rm = TRUE` scores them from that single answer, listwise deletion drops them. **83** answered neither and are unscoreable under any rule. With two items the available-case rule is a bigger leap than it was with four — half of a person's index can rest on one response — which is why the choice deserves a sentence in any write-up, not a silent default.

(e) The mean (0.51) and the median (0.50) all but coincide — the signature of a roughly **symmetric** distribution. Compare the lab's economic pair: identical pipeline, mean 0.90 pinned under a median of 1.00 at the ceiling. The histogram agrees — a centered hump using the whole scale, with real mass at *both* extremes:

```
ggplot(filter(voter, !is.na(social_index)), aes(x = social_index)) +
  geom_histogram(binwidth = 1/6, boundary = -1/12,
                fill = gov_colors[["carnelian"]], color = "white",
                linewidth = 0.2) +
  labs(x = "Social-issues importance index (0-1)", y = "Respondents",
       title = "Roughly symmetric: the country splits on social-issue importance") +
  theme_gov(base_size = 11)
```

Roughly symmetric: the country splits on social-issue importance



In 2012, with the economy in every headline, near-universal agreement that economic issues mattered — and genuine division over whether gay rights and abortion did — is exactly what the two shapes record. Same cleaning pipeline, opposite distributions: this is why you describe before you summarize.

Question 7: Don't know, but louder

Set up the three rules:

```
x_keep <- voter$important_gayr # Rule 1: careless
x_na   <- na_if(voter$important_gayr, 5) # Rule 2: DK -> NA
x_mid  <- if_else(voter$important_gayr == 5, 2.5, # Rule 3: DK -> 2.5
                  voter$important_gayr)
```

(a) The Don't knows and their share of nonmissing answers:

```
c(n_dk = sum(x_keep == 5, na.rm = TRUE),
  share = mean(x_keep == 5, na.rm = TRUE))
```

```
      n_dk      share
39.000000000 0.004941713
```

39 respondents, about **0.5%** of the item's 7,892 nonmissing answers.

(b) Mean and median under each rule:

```
tibble(
  rule   = c("Rule 1: keep the 5s", "Rule 2: DK to NA", "Rule 3: DK to 2.5"),
  mean   = c(mean(x_keep, na.rm = TRUE), mean(x_na, na.rm = TRUE),
              mean(x_mid, na.rm = TRUE)),
  median = c(median(x_keep, na.rm = TRUE), median(x_na, na.rm = TRUE),
              median(x_mid, na.rm = TRUE))
)
```

```
# A tibble: 3 x 3
```

```
  rule          mean median
<chr>      <dbl>  <dbl>
```

1 Rule 1: keep the 5s	2.69	3
2 Rule 2: DK to NA	2.68	3
3 Rule 3: DK to 2.5	2.68	3

(c) The **mean** moved under every rule change; the **median** sat at 3 throughout. The careless mean moved *upward* (2.688 versus the honest 2.676), because every kept 5 sits above the honest mean and drags it toward the top of the scale. The median is **resistant** (or **robust**): it depends only on the rank of the middle observation, and relocating 39 values out of roughly 7,900 cannot shift which value sits in the middle when nearly two thousand respondents occupy the center of the scale. (Rule 3 barely differs from Rule 2 here only because 2.5 happens to sit near this item’s honest mean — that is luck, not a justification; the midpoint rule *invents* an answer for people who explicitly declined to give one.)

(d) (Stretch.) With DK share p and honest mean m , keeping the 5s gives a weighted mean of the two groups — the $1 - p$ who answered, averaging m , and the p who said Don’t know, all sitting at 5:

```
p <- mean(x_keep == 5, na.rm = TRUE)
m <- mean(x_na, na.rm = TRUE)
c(identity = (1 - p) * m + 5 * p,
  direct   = mean(x_keep, na.rm = TRUE))
```

```
identity    direct
2.687658 2.687658
```

The identity reproduces the careless mean exactly. Two uses:

The hypothetical. Same honest mean, but $p = 0.20$:

```
0.8 * m + 5 * 0.2
```

```
[1] 3.14094
```

The careless mean would read **3.14** against an honest 2.68 — nearly half a point on a three-point-wide scale, from a rule change no output would ever flag. Our 0.5% DK share made carelessness cheap this time; nothing guarantees that on the next item.

The comparison. Rearranging, the distortion is $(1 - p)m + 5p - m = p(5 - m)$: it grows with the distance between the honest mean and the DK code. Check both items:

```
c(econ_careless = mean(voter$important_econ, na.rm = TRUE),
  econ_honest   = mean(na_if(voter$important_econ, 5), na.rm = TRUE))
```

```
econ_careless  econ_honest
1.140200      1.121058
```

`important_econ` moves by about 0.019 and `important_gayr` by about 0.011 — same 39 Don’t knows, nearly double the damage — because the economy item’s honest mean (1.12) sits much farther from 5 than the gay-rights item’s (2.68): $p(5 - 1.12) > p(5 - 2.68)$. The items *most* skewed toward “important” are exactly the ones a disguised code distorts most.

Question 8: One good graph

Many geometries work here; this is one of them. Because the party groups differ in size (3,607 scored Democrats versus 1,168 Others), the y-axis shows the share *within* each party, not counts:

```

party_dist <- voter |>
  mutate(party = factor(pid3, 1:3, c("Democrat", "Republican", "Other"))) |>
  filter(!is.na(party), !is.na(social_index)) |>
  count(party, social_index) |>
  group_by(party) |>
  mutate(share = n / sum(n))

ggplot(party_dist, aes(x = social_index, y = share)) +
  geom_col(fill = gov_colors[["carnelian"]], width = 0.13) +
  facet_wrap(~party, nrow = 1) +
  scale_y_continuous(labels = scales::label_percent(accuracy = 1)) +
  scale_x_continuous(breaks = c(0, 0.5, 1)) +
  labs(x = "Social-issues importance index (0-1)", y = "Share of party group",
       title = "Democrats rated social issues as more important in 2012") +
  theme_gov(base_size = 11)

```

Democrats rated social issues as more important in 2012

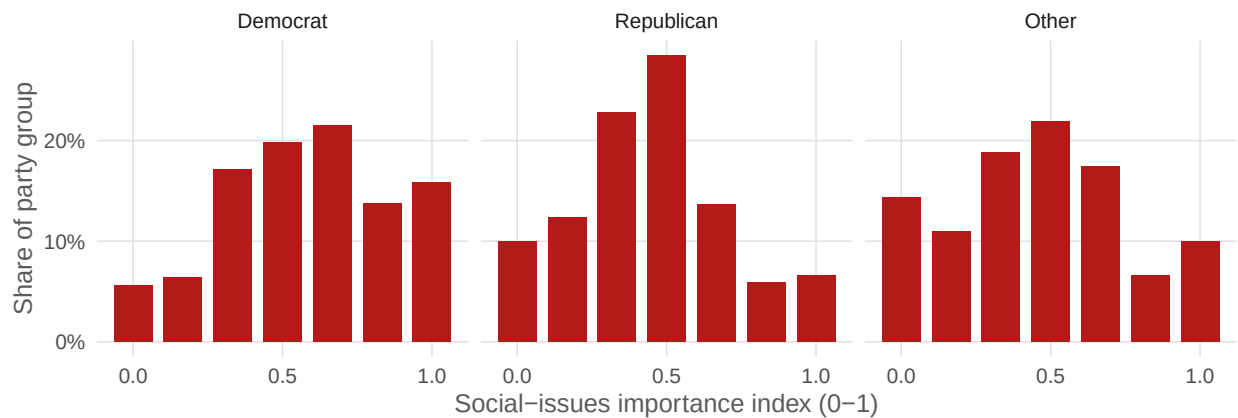


Figure 1: Democrats sat visibly higher on the social-issues importance scale in 2012 — half score in the top third of the index, against about a quarter of Republicans — the mirror image of the economic index, where the parties’ positions reverse. Shares describe the 7,836 scored respondents who reported a party identification (81 scored respondents skipped the party question), computed within each party group because the groups differ in size.

What the checklist looked like in practice:

- **Labeled factors:** `factor(pid3, 1:3, c("Democrat", ...))` — a panel headed “1” communicates nothing.
- **Shares, not counts:** raw counts would make the Republican panel tower over the Other panel only because scored Republicans outnumber scored Others by well over two and a half to one (3,061 versus 1,168).
- **Zero baseline:** these are bars, so the y-axis starts at 0 — the bar’s length *is* the datum.
- **Title as message, caption as finding plus scope:** the second caption sentence says exactly whose data the shares describe — including who was excluded and why; whether the pattern holds beyond these respondents is, as always this week, a question we are not yet equipped to

answer.

- **One message:** the figure shows the party gap in one variable. Age, education, and everything else belong in their own figures.

A grouped **boxplot** was a perfectly acceptable geometry *here*, because $\hat{Q}_{0.25}$, $\hat{Q}_{0.5}$ and $\hat{Q}_{0.75}$ genuinely differ across the scale:

```
voter |>
  mutate(party = factor(pid3, 1:3, c("Democrat", "Republican", "Other"))) |>
  filter(!is.na(party), !is.na(social_index)) |>
  group_by(party) |>
  summarise(q25 = quantile(social_index, 0.25),
            q50 = quantile(social_index, 0.50),
            q75 = quantile(social_index, 0.75))
```

```
# A tibble: 3 x 4
  party      q25  q50  q75
<fct>    <dbl> <dbl> <dbl>
1 Democrat 0.333 0.667 0.833
2 Republican 0.333 0.5   0.667
3 Other    0.167 0.5   0.667
```

The Democratic box sits visibly above the other two (median 0.67 versus 0.50). But try the same geometry on the lab's *economic* index and it fails: for Republicans $\hat{Q}_{0.25}$, $\hat{Q}_{0.5}$ and $\hat{Q}_{0.75}$ are all 1.0 there, so the box collapses to a line at the ceiling and hides the entire distribution it is meant to summarize. The rule to take away: a boxplot is a summary of quartiles, so check that the quartiles differ before you let five numbers stand in for eight thousand.